Mentora AI Concept Note

IBM SkillsBuild for Academia Internship Project

SDG 4 Quality Education | Agentic AI Learning Platform | October 2026

Project summary. Mentora AI is a working web prototype that personalizes learning around a student's career goal, prior skills, assessment results, and ongoing performance. The system combines specialist AI agents, grounded retrieval, adaptive quizzes, career guidance, project recommendations, reminders, and progress tracking in one learner experience.

Problem and opportunity

Students often move between videos, course platforms, search engines, and general chat tools without a coherent learning plan. Static courses rarely adjust to career goals or changing performance. New AI developments also make learning material age quickly, while limited mentoring and weak accountability reduce completion. These gaps directly affect SDG 4 because access alone does not guarantee relevant, effective, and inclusive learning.

- Learning resources remain scattered across unrelated platforms.
- Students receive limited feedback on weak topics and next steps.
- Career advice, portfolio projects, and learning progress are usually disconnected.
- Generic chat responses can lack grounding, learner context, and persistent evidence.

Proposed solution

Mentora AI provides a single learner workspace. A student creates a profile, chooses a career goal, completes an initial assessment, and receives a sequenced roadmap. The platform records progress and quiz evidence, routes requests to specialist agents, retrieves relevant material from a curated knowledge base, and recommends skill gaps and portfolio projects.

Capability	Learner value	Current prototype
Personalization	A plan shaped by goals and evidence	Profile, assessment bands, adaptive roadmap
Teaching	Contextual explanations with traceable sources	Tutor routing with local ChromaDB retrieval
Assessment	Difficulty changes with performance	Persisted quizzes, explanations, weak-topic updates
Career support	Clear skills and portfolio direction	Skill gaps, certifications, interview preparation, ranked projects
Engagement	Visible consistency and follow-up	Activity history, streaks, reminders, dashboard

Agentic system design

The orchestrator classifies each request and selects a specialist. The Tutor explains concepts using retrieved knowledge. Quiz Master directs students to scored adaptive quizzes. Career Coach uses the learner profile to explain skill gaps. Project Guide recommends feasible portfolio work. Researcher presents curated AI updates. Reminder Agent helps the learner create an actionable follow-up.

- React and TypeScript provide the responsive learner interface.
- FastAPI exposes authenticated APIs and orchestrates domain services.
- SQLAlchemy and Alembic manage learner records, assessments, roadmaps, quizzes, projects, activities, and reminders.
- ChromaDB stores local knowledge chunks with deterministic offline embeddings.
- A provider boundary supports a no-key deterministic demo and optional OpenAI responses.

Implemented learner journey

Step	Stage	Outcome
1	Profile	Record education, experience, skills, career goal, and weekly study time.
2	Assessment	Measure topic performance and classify strong, average, and weak areas.
3	Roadmap	Create an ordered path with locked progression and backend-owned completion.
4	Learn and practise	Ask a grounded mentor and complete adaptive, scored quizzes.
5	Career evidence	Review skill gaps, projects, certifications, progress, and activity streaks.

Target users and adoption

Primary users are college and engineering students, including learners entering AI from non-technical backgrounds. Early adopters include AI clubs, IBM SkillsBuild learners, colleges with AI labs, and students preparing for AI careers. Institutions, CSR education programs, and government skill initiatives can support broader adoption after a validated pilot.

Validation and evidence

The current submission includes an operational local MVP rather than a concept-only mock-up. Automated verification covers authentication, user isolation, assessment and roadmap personalization, adaptive quizzes, retrieval, agent routing, career analysis, projects, reminders, activity, and streaks.

Verification	Result
Backend automated tests	12 passed

Verification	Result
Frontend unit tests	3 passed
Frontend quality checks	Lint, TypeScript, and production build passed
Runtime smoke test	Authenticated core routes and grounded mentor response passed

Impact and success measures

A pilot should evaluate whether Mentora helps learners complete planned topics, improve quiz performance, return consistently, and produce career evidence. The first evaluation should track roadmap completion, quiz improvement by topic, active learning days, project progress, and learner feedback. Institutional adoption and reminder engagement become relevant only after those channels exist.

Responsible scope and limitations

The prototype keeps offline behavior honest. It uses curated Markdown knowledge, local retrieval, deterministic personalization, and in-app reminders. Research updates are curated records. Live OpenAI mode requires a configured key and separate quality evaluation. WhatsApp, email, automated crawling, native mobile, LMS, payments, and marketplaces remain future integrations. Mentora AI is an internship project and is not presented as an official IBM product.

Implementation plan

- Pilot the current web MVP with a small student cohort and collect structured feedback.
- Measure retrieval quality and learning outcomes before expanding the knowledge corpus.
- Add monitored live-model evaluation, production authentication hardening, PostgreSQL, backups, and shared rate limiting before public deployment.
- Assess consent, privacy, and message-delivery requirements before enabling email or WhatsApp reminders.

Conclusion

Mentora AI turns the IBM SkillsBuild example into a tested prototype centered on learner evidence. The product demonstrates how agent routing, retrieval, adaptive assessment, and career guidance can work together while keeping future commercial integrations clearly separated from the current MVP.